REHOTEQ

AI EDUCATION CURRICULUM

Comprehensive Teacher's Lesson Note

Algorithms

Junior Secondary 1 · Third Term · Week 4

Field	Detail
Subject	AI Education and Digital Skills
Academic session	2026/2027
Duration	2 periods (40 to 45 minutes each, subject to the school timetable)
Document type	Comprehensive Teacher's Lesson Note
Curriculum status	Proposed curriculum for school pilot, review and consultation
Prepared by	Rehoteq Technologies (rehoteq.com)
Edition	First edition, November 2026

PROPOSED CURRICULUM NOTICE: This lesson note expands the Rehoteq AI Education Curriculum's proposed Junior Secondary 1, Third Term, Week 4 scheme of work. It is offered for school pilot, review and consultation. It is not represented as an officially approved national syllabus, and it does not replace the current NERDC curriculum or any examination-board requirement.

1. Lesson Information

Field	Detail
Class	JSS 1
Subject	AI Education and Digital Skills
Term	Third Term
Week	Week 4 of Third Term
Topic	Algorithms: definition and everyday examples; the cooking algorithm
Duration	2 periods (40 to 45 minutes each)

2. Learning Outcome

By the end of the lesson, learners can:

- Define an algorithm as a precise step-by-step set of instructions to finish a task.
- Give everyday examples of algorithms and spot what makes one work.
- Write the algorithm for a familiar cooking task in clear ordered steps.
- Test a classmate's algorithm by following it exactly and reporting what breaks.

3. Weekly Scheme Content

Period	Scheme content
Period 1	Definition; everyday examples
Period 2	Write the algorithm for cooking a meal
Weeks 11 to 13 of the term are reserved for examinations, marking and results. Do not introduce new content during those weeks.	

4. Materials and Preparation

- A simple recipe card (a familiar local dish, no brand names), one per pair; paper and pencils.
- Instruction cards for the robot-teacher game: one learner is the robot who obeys instructions exactly.
- The class's flowchart vocabulary is next week; this week stays with numbered steps.

No-lab fallback: the whole week is unplugged. The robot-teacher game and the cooking algorithm need only paper and a willing robot.

5. Period One: Introduce and Model

- Starter (5 minutes): the robot-teacher game. One learner is the robot; the class gives instructions to make the robot walk from the door to their seat. Every ambiguous instruction is obeyed literally, and the class laughs and fixes it.
- Teach (10 minutes): an algorithm is a precise, ordered set of steps that finishes a task. Precise means no guessing: each step says exactly what to do. Everyday algorithms: the steps for washing hands, for boarding a bus, for charging a phone.
- Spot and sort (15 minutes): teams put three everyday routines in order and mark any step that is vague (add some water: how much?).

6. Lesson Pictures

Show these pictures while teaching. Use each one to point back at the week it comes from before the class moves on.



Picture 1. The robot obeys every word: say exactly what to do.



Picture 2. Every day runs on algorithms: hands, buses, phones.



Picture 3. Vague steps break algorithms: how much water, exactly?



Picture 4. Write the meal algorithm, then follow it exactly.

7. Period Two: Practise and Apply

- Cooking algorithm (25 minutes): pairs write the algorithm for cooking a simple familiar meal: exact quantities (three cups), exact actions (stir until it thickens), exact order. The teacher supplies the known recipe as a reference.
- Swap and test (15 minutes): pairs swap algorithms and follow them exactly as written; any step that fails or confuses is marked. Pairs get their algorithms back and fix the marked steps.
- Close (5 minutes): the class agrees on what makes an algorithm good: precise, ordered, and finishable by someone who has never seen the task.

8. Assessment and Evidence

- Evidence to collect: each pair's cooking algorithm (written and corrected) and one test note from the swapping pair.

Success looks like: algorithms are numbered, precise (no 'some' or 'a bit'), and survive a strict follow-through with at most small fixes.

9. Suggested Questions and Answer Guidance

1. What makes an instruction precise? Expected response: it says exactly what to do, with amounts and order, so no guessing is needed.

2. Why does order matter? Expected response: some steps depend on earlier ones; the wrong order changes or spoils the result.

3. Give one everyday algorithm and its first step. Expected response: any sensible routine with a sensible first step.

4. How is an algorithm different from a program? Expected response: an algorithm is the plan; a program is the plan written in a language a computer can run.

10. Home Activity

Learners write the algorithm for one home task in five steps (sweeping the compound, washing a plate, fetching water) and bring it; no device needed.

11. Inclusion

Instructions may be given, written or signed; the robot obeys any clear signal the class agrees on, including gestures. Reading aloud is shared so no learner is put on the spot alone. No colour-only cues in any card.

12. Safety and Privacy

Cooking stays on paper: no heat, knives or real food handling. Robot-walking routes are checked for obstacles before the game starts.

13. Teacher Reflection

After the lesson, note what worked, which parts confused the class, which pupils need follow-up, and one change for next term.

Reflection notes

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14. Curriculum Alignment and Document Status

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